本章提纲

1 环境配置

1.1 前期准备

1.1.1 安装IDE

- 安装**Python IDE**（集成开发环境），推荐 PyCharm，已经提供下载好文件，双击安装

- PyCharm 是由 JetBra pycharm-community-2020.2.3.exe PyCharm 具备一般 Python IDE 的功能，比如：调试、语法高亮、项目管理、代码跳转、智能提示、自动完成、单元测试、版本控制等。

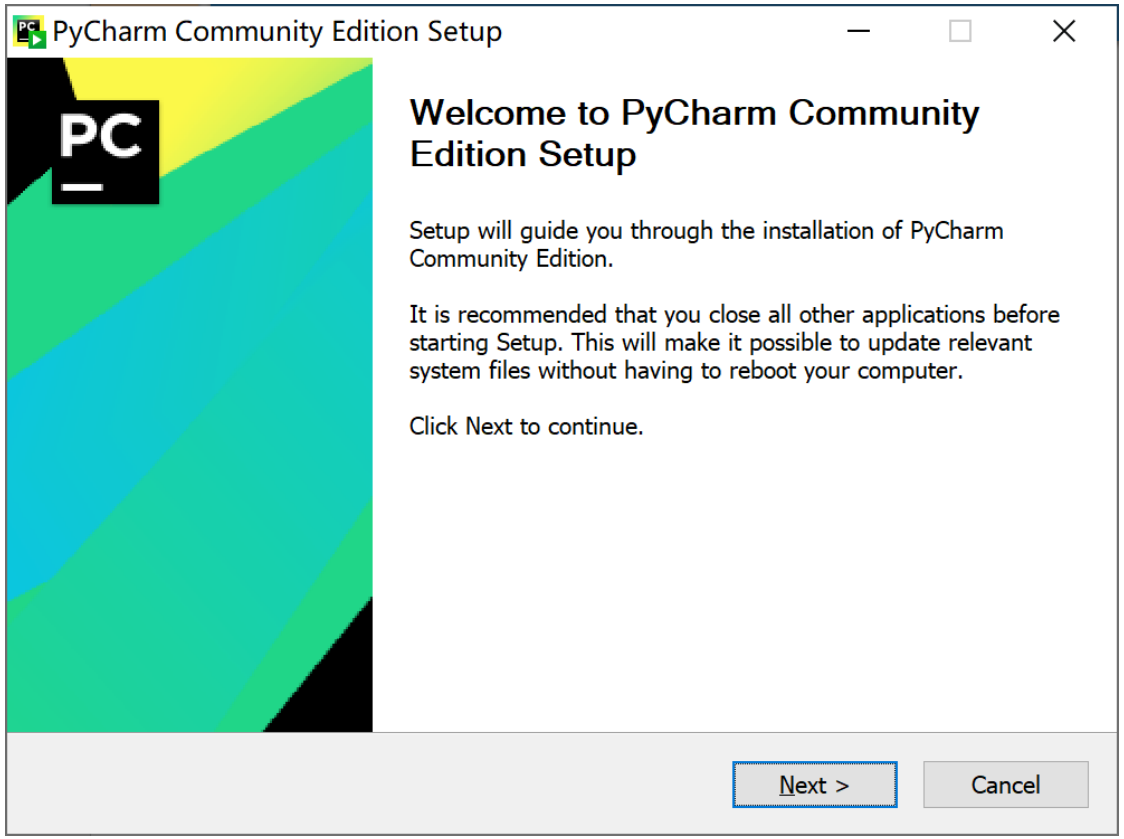
- 其他可选IDE：**Vscode、 Jupyter**

- 注：如果已经拥有某一软件，可以完成个性化安装，可以跳过步骤，进行独立作业。

1.1前期准备

1.1.1 安装pycharm

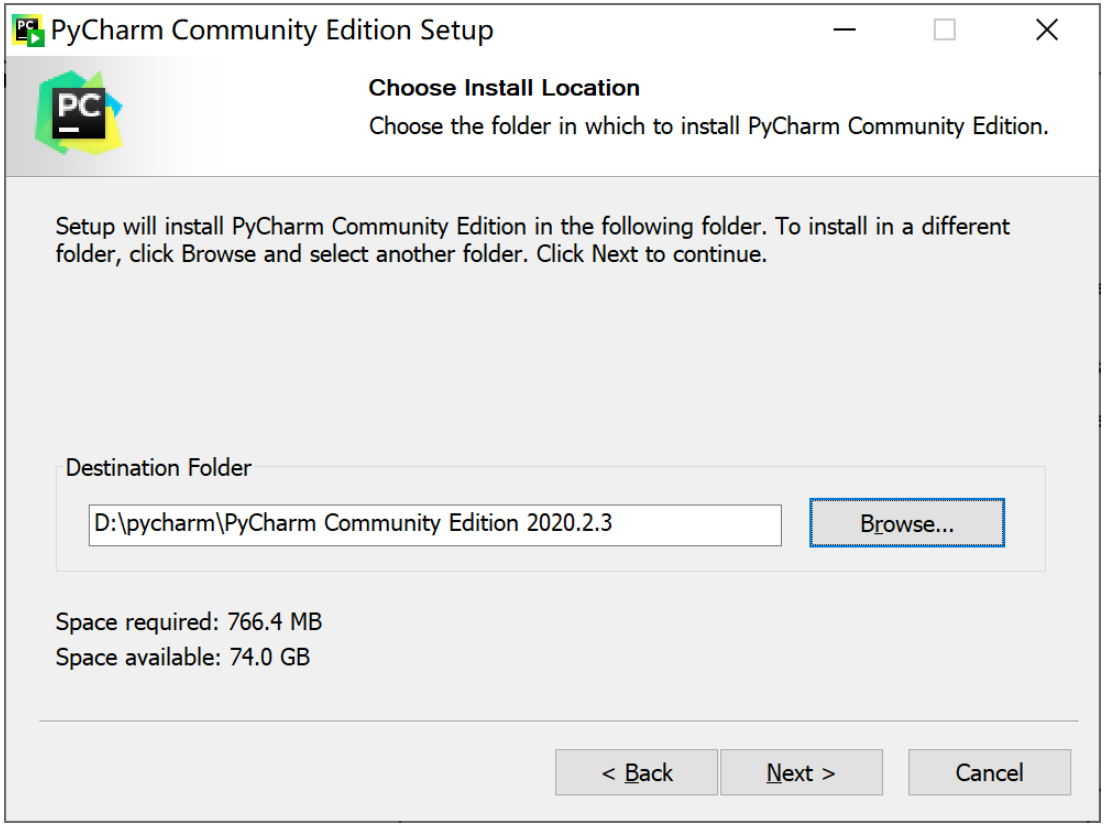
- 点击next



1.1前期准备

1.1.1 安装pycharm

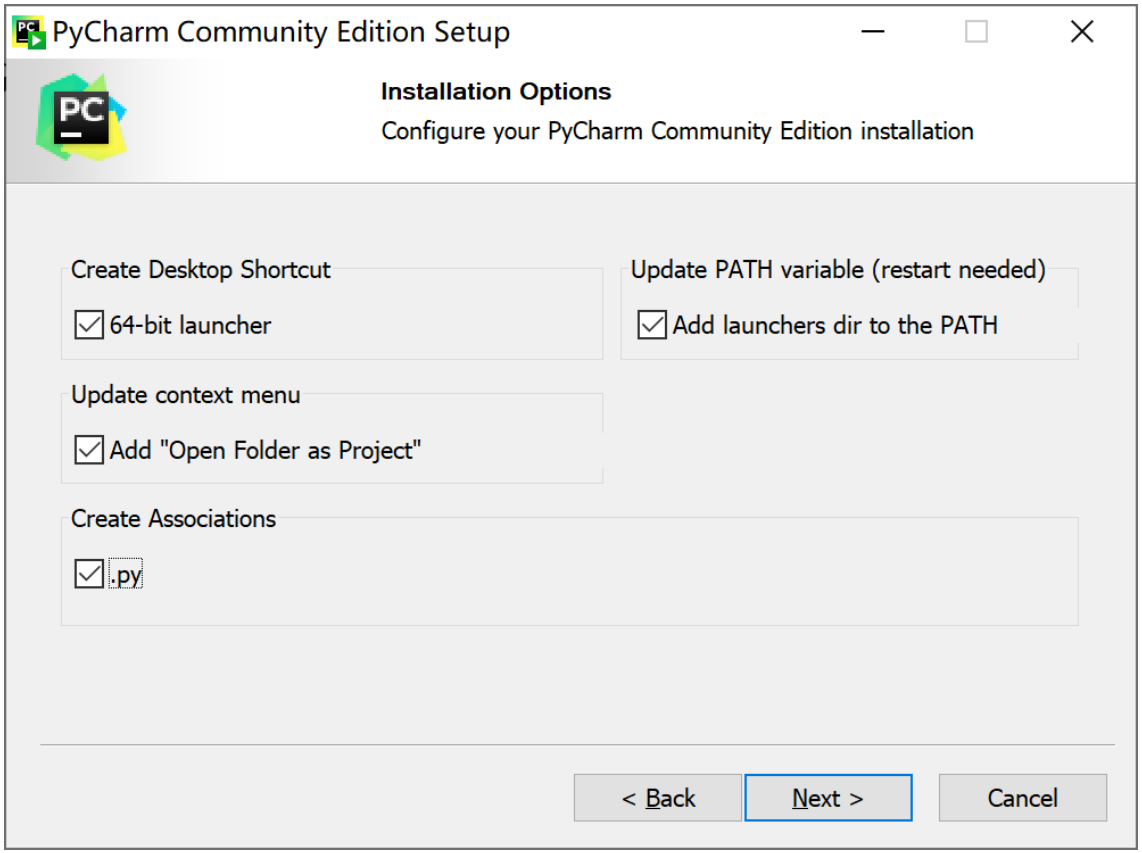
- （可选）点击Browse修改安装路径
- 点击Next



1.1前期准备

1.1.1 安装pycharm

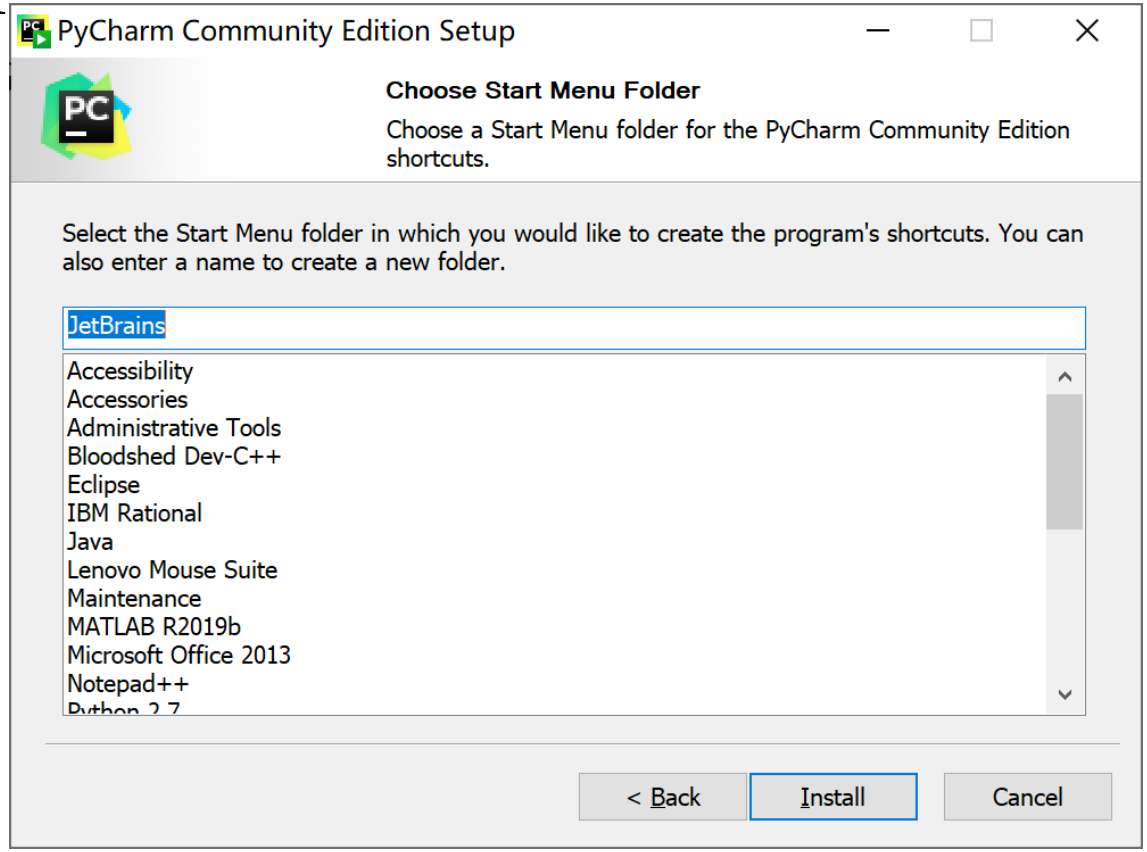
- 全部勾选
- 点击next



1.1前期准备

1.1.1 安装pycharm

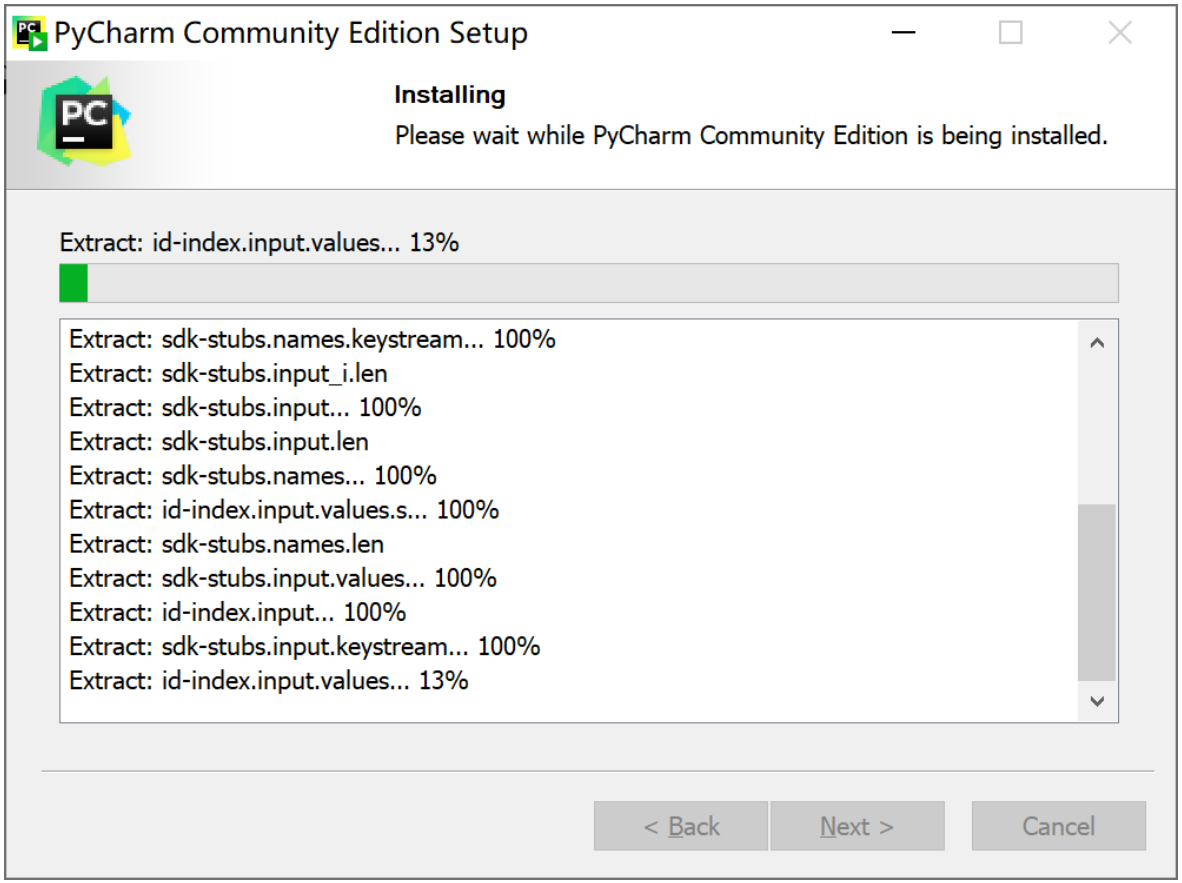
- 点击Install



1.1前期准备

1.1.1 安装pycharm

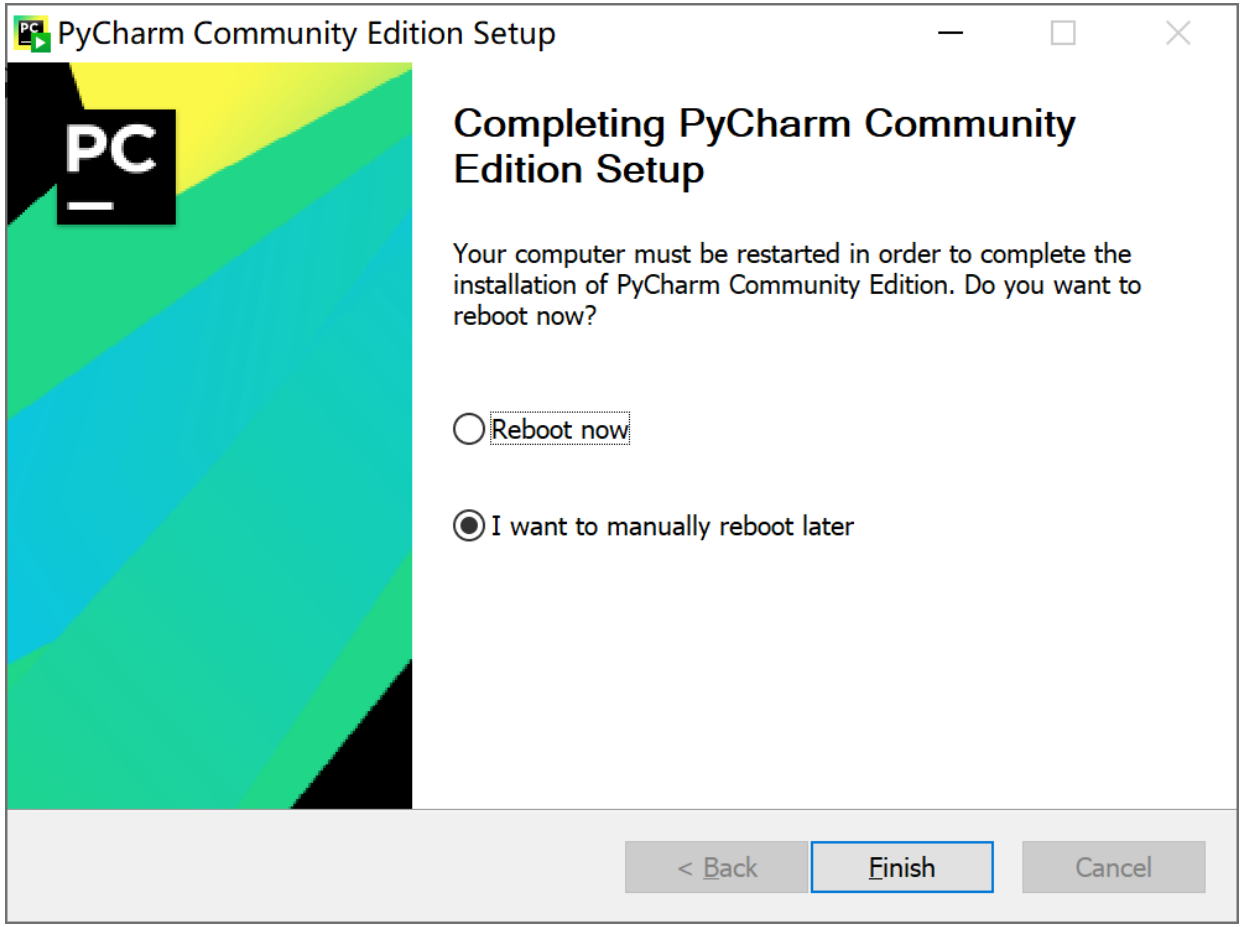
➢等待完成



1.1前期准备

1.1.1 安装pycharm

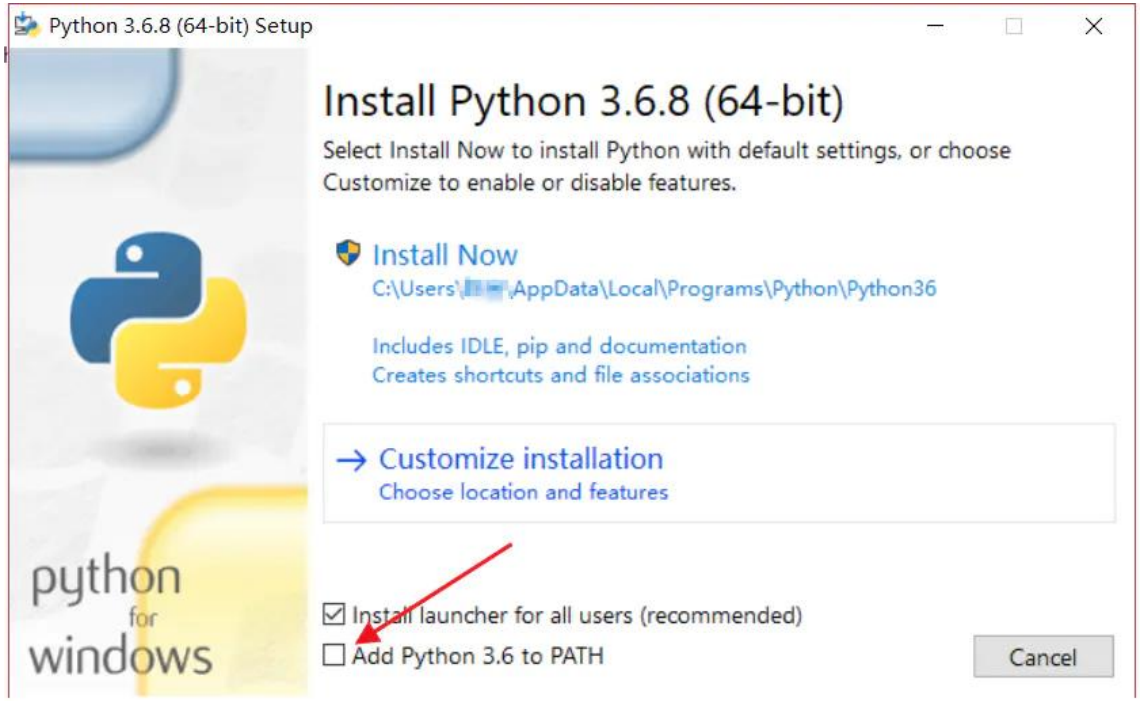
- 稍后重启
- 点击Finish



1.1前期准备

1. 1. 2 安装python

- 勾选添加路径
- 第一个为默认安装(选择默认安装)
- 第二个为自定义安装



1.2 Anaconda下载

1. 2. 1 下载

在机器学习，深度学习中，要用到大量的 package。如果说，函数是一个工具，那么 package 就是一个工具包。一个个安装 package 很麻烦，而且容易出现疏漏。于是，就有了 Anaconda，这是一个集成了常用于科学分析的大量 package。也就是说，你只要安装了 Anaconda，就安装了很多我们之后要用的许多packages。

Anaconda 的下载地址：
https://www.anaconda.com/products/individual
Anaconda历史版本链接：
https://repo.anaconda.com/archive/
清华源下载链接：
https://mirrors.tuna.tsinghua.edu.cn/anaconda/

https://mirrors.tuna.tsinghua.edu.cn/anaconda/archive/

网 前人栽树 学术造假 经验贴 哔哩哔哩 (°- °)つロ ... ChatGPT 网易邮箱 csapp

tuna 清华大学开源软件镜像站

HOME EVENTS BLOG RSS PODCAST

Index of /anaconda/archive/

Last Update: 202

File Name ↓	File Size ↓	Date ↓
Parent directory/	-	-
Anaconda-1.4.0-Linux-x86.sh	220.5 MiB	2013-07-04 01:47
Anaconda-1.4.0-Linux-x86_64.sh	286.9 MiB	2013-07-04 17:26
Anaconda-1.4.0-MacOSX-x86_64.sh	156.4 MiB	2013-07-04 17:40
Anaconda-1.4.0-Windows-x86.exe	210.1 MiB	2013-07-04 17:48
Anaconda-1.4.0-Windows-x86_64.exe	241.4 MiB	2013-07-04 17:58
Anaconda-1.5.0-Linux-x86.sh	238.8 MiB	2013-07-04 18:10
Anaconda-1.5.0-Linux-x86_64.sh	306.7 MiB	2013-07-04 18:22
Anaconda-1.5.0-MacOSX-x86_64.sh	166.2 MiB	2013-07-04 18:37
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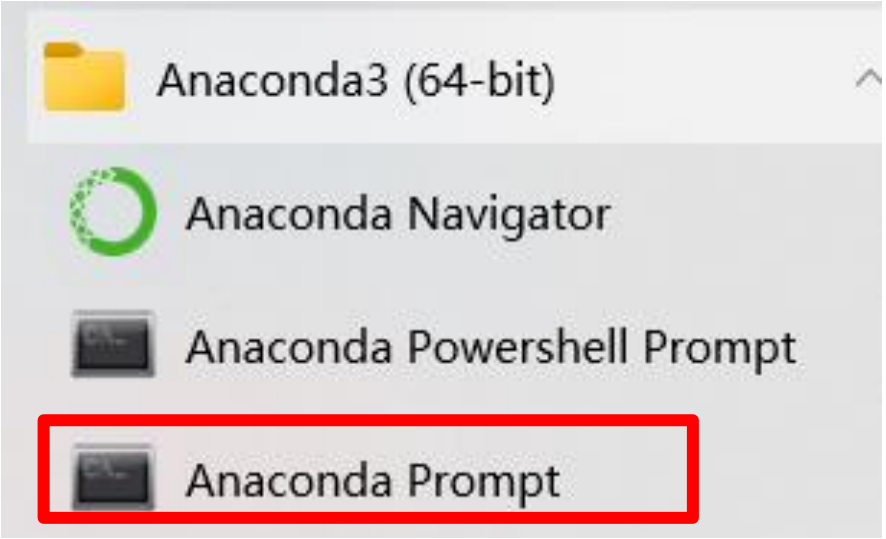
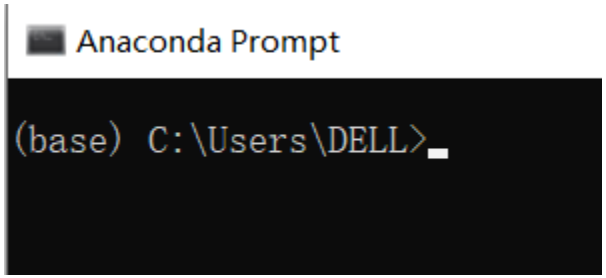
Anaconda-1.6.0-Linux-x86.sh	241.6 MiB	2013-07-04 19:19
Anaconda-1.6.0-Linux-x86_64.sh	309.5 MiB	2013-07-04 19:32
Anaconda-1.6.0-MacOSX-x86_64.sh	169.0 MiB	2013-07-04 19:47
Anaconda-1.6.0-Windows-x86.exe	244.9 MiB	2013-07-04 19:56

1.2 Anaconda下载

1.2.2 安装成功





1.2 Anaconda下载

1. 2. 2 安装成功

常用命令

- 对anaconda prompt 界面进行清屏 cls
- 查看已经安装的环境 conda info --envs
- 创建一个新的环境（名字为tensorflow2） conda create -n tensorflow2
- 删除一个已有的环境（名字为tensorflow2）conda env remove –n tensorflow2
- 进入到某个创建的环境中（进入到tensorflow2环境中）activate tensorflow2

1.2 Anaconda下载

1.2.3 新建pytorch环境

- conda create -n pytorch python=3.n （根据自己的py版本）

```
(base) C:\Users\DELL>conda create -n pytorch python=3.6
Retrieving notices: ...working... done
WARNING: A conda environment already exists at 'D:\Env\Anaconda3\envs\pytorch'
Remove existing environment (y/[n])? y

Collecting package metadata (current_repodata.json): done
Solving environment: done
```

```
The following NEW packages will be INSTALLED:

  certifi            pkgs/main/noarch::certifi-2020.6.20-pyhd3eb1b0_3
  pip                conda-forge/win-64::pip-20.0.2-py36_1
  python             pkgs/main/win-64::python-3.6.13-h3758d61_0
  python_abi         conda-forge/win-64::python_abi-3.6-2_cp36m
  setuptools         conda-forge/win-64::setuptools-49.6.0-py36ha15d459_3
  sqlite             pkgs/main/win-64::sqlite-3.41.2-h2bbff1b_0
  vc                 pkgs/main/win-64::vc-14.2-h21ff451_1
  vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
  wheel              pkgs/main/noarch::wheel-0.37.1-pyhd3eb1b0_0
  wincertstore       pkgs/main/win-64::wincertstore-0.2-py36h7fe50ca_0


Proceed ([y]/n)? y
```

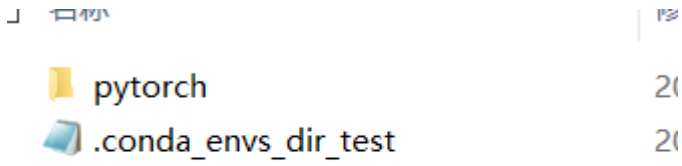
1.2 Anaconda下载

1.2.3 新建pytorch环境

- conda activate pytorch

```
(base) C:\Users\DELL>conda activate pytorch

(pytorch) C:\Users\DELL>
```



1.3 pytorch下载

1. 3. 1 下载

官网 https://pytorch.org/get-started/locally/

recommended package manager since it installs all dependencies. You can also install previous versions of PyTorch. Note that LibTorch is only available for C++.

NOTE: Latest PyTorch requires Python 3.8 or later. For more details, see Python section below.

PyTorch Build	Stable (2.2.1)		Preview (Nightly)	
Your OS	Linux	Mac	Windows	
Package	Conda	Pip	LibTorch	Source
Language	Python		C++ / Java	
Compute Platform	CUDA 11.8	CUDA 12.1	~~ROCm 5.7~~	CPU
Run this Command:	`conda install pytorch torchvision torchaudio pytorch-cuda=11.8 -c pytorch -c nvidia`			

寻找和自己python版本对应的pytorch版本

查看对应的pytorch https://blog.csdn.net/WOSHIRENXIN/article/details/127415609

1.3 pytorch下载

1.3.1 下载

```
Proceed ([y]/n)? y


Downloading and Extracting Packages
pillow-8.3.1          | 676 KB    | ##################################################################################### | 100%
xz-5.4.5              | 593 KB    | ##################################################################################### | 100%
mkl-service-2.3.0     | 45 KB     | ##################################################################################### | 100%
mkl-2020.2            | 109.3 MB  | ##################################################################################### | 100%
numpy-base-1.19.2     | 3.8 MB    | ##################################################################################### | 100%
mkl_fft-1.3.0         | 131 KB    | ##################################################################################### | 100%
intel-openmp-2023.1.  | 2.7 MB    | ##################################################################################### | 100%
typing_extensions-4.  | 28 KB     | ##################################################################################### | 100%
numpy-1.19.2          | 22 KB     | ##################################################################################### | 100%
torchvision-0.11.0    | 8.5 MB    | ##################################################################################### | 100%
dataclasses-0.8       | 22 KB     | ##################################################################################### | 100%
pytorch-1.10.0        | 951.0 MB  | ###3                                                                                |   4%
olefile-0.46          | 49 KB     | ##################################################################################### | 100%
torchaudio-0.10.0     | 2.1 MB    | ##################################################################################### | 100%
certifi-2021.5.30     | 140 KB    | ##################################################################################### | 100%
cudatoolkit-10.2.89   | 317.2 MB  | ########################7                                                            |  33%
mkl_random-1.1.1      | 235 KB    | ##################################################################################### | 100%

done

(pytorch) C:\Users\DELL>
(pytorch) C:\Users\DELL>
(pytorch) C:\Users\DELL>
(pytorch) C:\Users\DELL>
(pytorch) C:\Users\DELL>q
```

出现done代表完成安装

1.3 pytorch下载

1.3.1 下载

清华镜像源

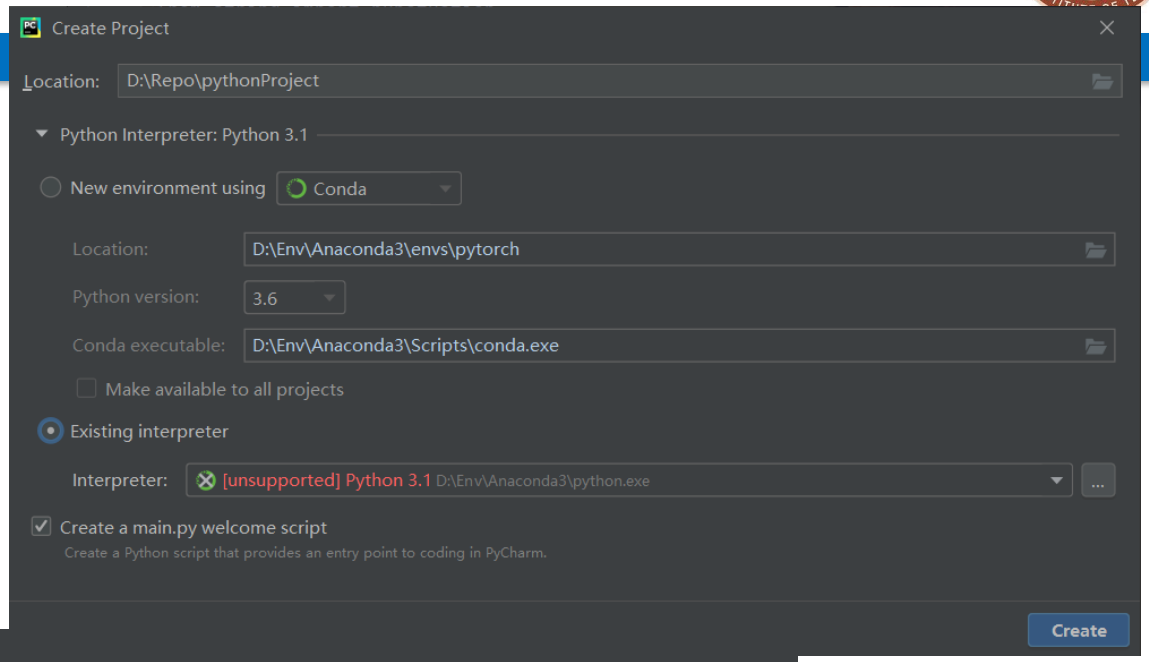
在安装过程中，我们会遇到很多问题，很多需要下载的包可能会由于网络连接的关系下载失败，因此，我们需要为Anaconda添加国内的镜像源，如清华的镜像源。添加镜像源的命令如下：

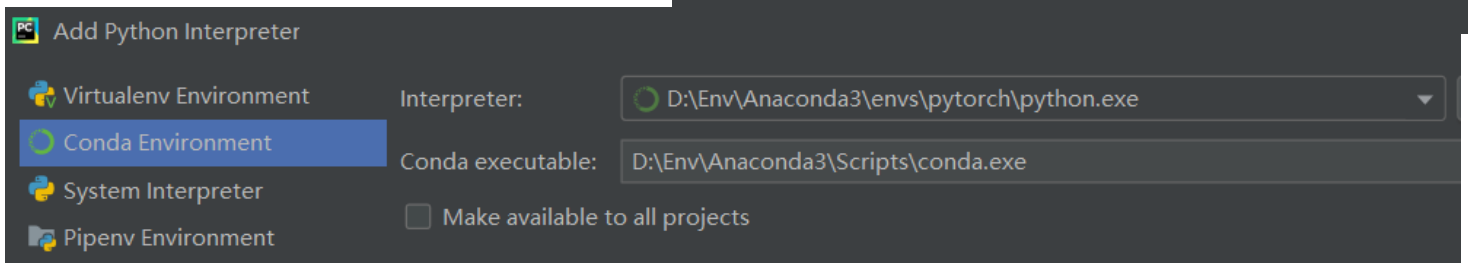
- conda config --add channels https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/free/
- conda config --add channels https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main/
- conda config --set show_channel_urls yes
- conda config --add channels https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/pytorch/

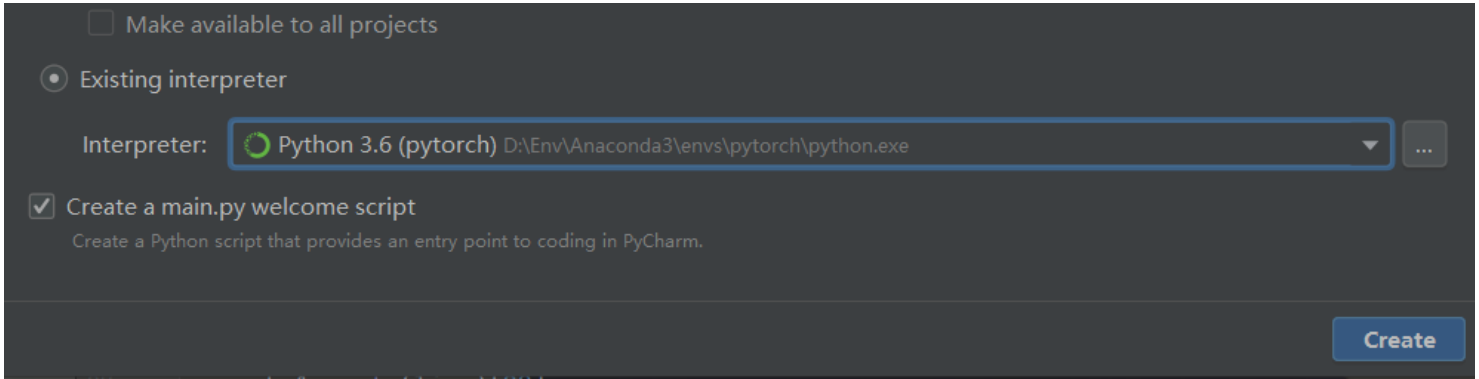
添加完镜像源后就可以执行官网给出的安装命令了。

1.4 py文件创建

1.4.1 新建项目

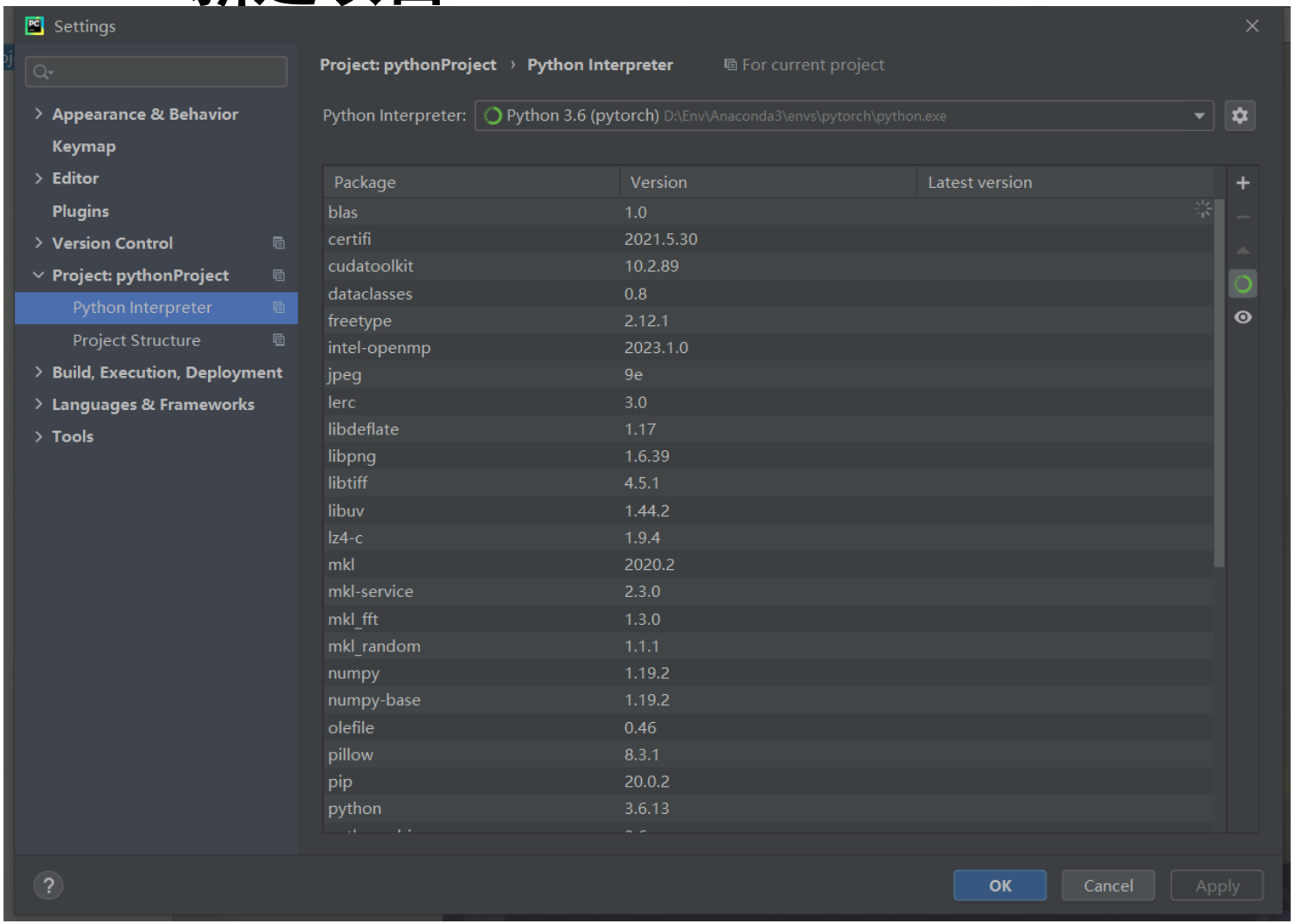




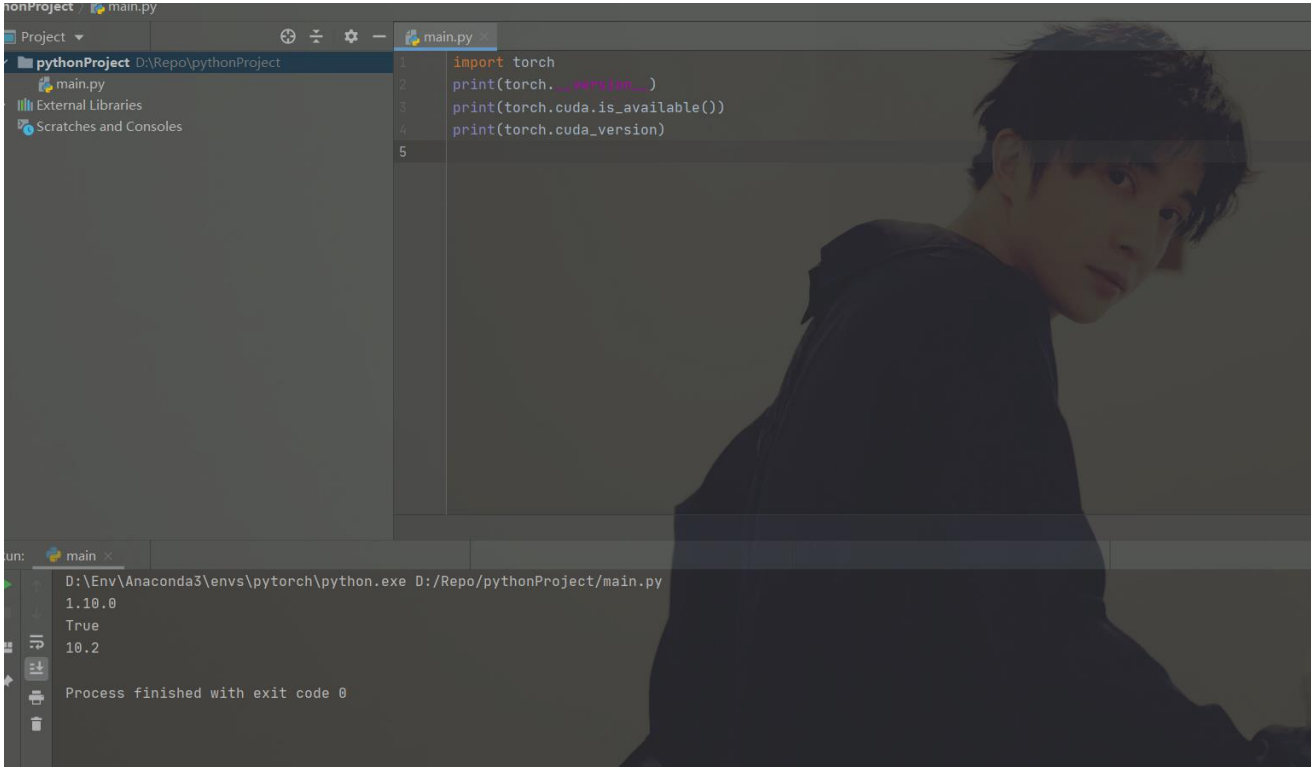


1.4 py文件创建

1.4.1 新建项目



1.4 py文件创建

1.4.2 测试代码

```
import torch
print(torch.__version__)
print(torch.cuda.is_available())
print(torch.cuda_version)
```